

you are a expert statistics teacher.create a details theoretical note about Quantiles and Percentiles , 5 number summary and BoxPlot and standard deviationin 3 seperate module.add a case study in business scenario about the interp

When studying data, we want to know not only where values tend to gather, but also where individual records stand and how widely spread they are.

Module 1: Quantiles and Percentiles

The Narrative: The Marathon Finish Line

Imagine you are standing at the finish line of a marathon with 10,000 runners. If someone asks, "Did runner #42 do well?", knowing their raw time of 3 hours and 45 minutes doesn't tell the full story without context.

If it was a rainy day with strong winds, maybe only a tiny fraction of runners finished faster. To evaluate their performance fairly, the race organizers line everyone up in order of their finish times. Once everyone is in order, the judges divide the entire field into equal segments to see which bracket each runner lands in.

Modern Practical Explanation

Quantiles are cut-points that divide an ordered dataset into continuous intervals with equal probabilities or frequencies.

Percentiles: Cut the ordered data into 100 equal parts. For instance, the **90th percentile** (P_{90}) means 90% of the observations fall at or below this value, while 10% fall above it.

Deciles: Cut the data into 10 equal parts (D_1 to D_9).

Quartiles: Cut the data into 4 equal quarters (Q_1 , Q_2 , Q_3).

Q_1 (**First Quartile / 25th Percentile**): 25% of the data lies below it.

Q_2 (**Second Quartile / 50th Percentile**): The **Median**, splitting the distribution exactly in half.

Q_3 (**Third Quartile / 75th Percentile**): 75% of the data lies below it.

Percentiles are robust against extreme outliers. Unlike averages, they tell you your relative standing regardless of how far the extreme values stretch.

Module 2: The Five-Number Summary and Box Plots

The Narrative: The Warehouse Cargo Inspector

Picture a quality assurance inspector auditing thousands of shipment crates arriving at a seaport. Weighing every single crate manually takes too much time, and calculating just the average weight can be misleading if a single crate is packed with solid lead.

Instead, the inspector picks five landmarks: the lightest standard crate, the lower cutoff where smaller shipments end, the true middle crate, the upper cutoff where heavy crates begin, and the heaviest crate that isn't considered an anomaly. With just these five numbers, the inspector sketches a quick diagram that shows the entire spread and identifies any unusual cargo.

Modern Practical Explanation

The **Five-Number Summary** provides a non-parametric summary of a distribution's center, spread, and range:

Minimum: The lowest value within regular data boundaries.

Q_1 : The median of the lower half of the data.

Median (Q_2): The center value of the dataset.

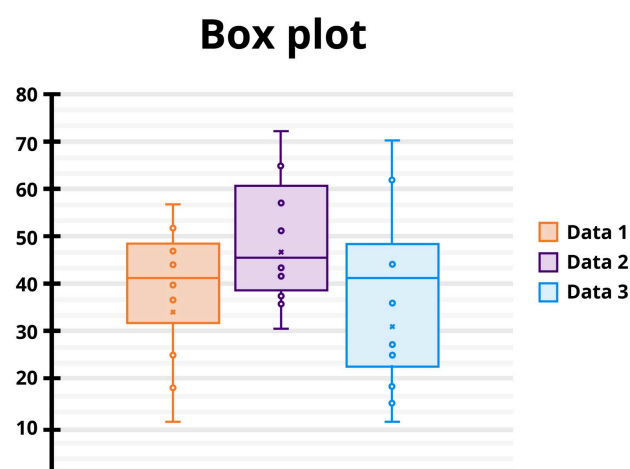
Q_3 : The median of the upper half of the data.

Maximum: The highest value within regular data boundaries.

The difference between Q_3 and Q_1 is called the **Interquartile Range (IQR)**:

$$\text{IQR} = Q_3 - Q_1$$

The IQR represents the middle 50% of your data and serves as the foundation for the **Box Plot** (Box-and-Whisker plot):



Anatomy of comparative box plots. Source: petrroudhny / Getty Images

The Box: Stretches from Q_1 to Q_3 , capturing the central 50% of the distribution.

The Center Line: Marks the median (Q_2).

Whiskers: Extend from the box edges to the smallest and largest values that sit within $1.5 \times \text{IQR}$ from the hinges ($Q_1 - 1.5 \times \text{IQR}$ and $Q_3 + 1.5 \times \text{IQR}$).

Outliers: Individual points plotted beyond the whiskers.

Module 3: Standard Deviation

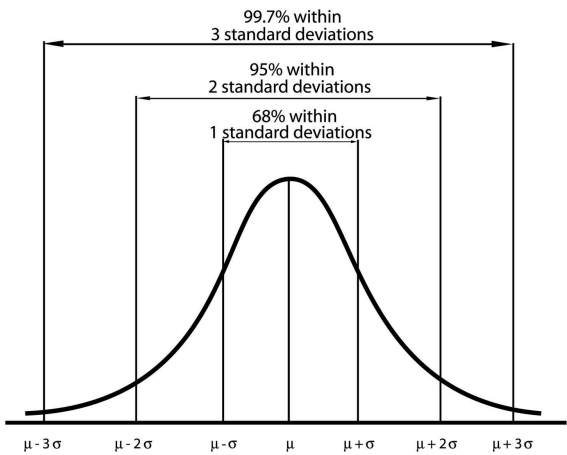
The Narrative: The Two Coffee Roasters

Imagine two local cafes, Roaster A and Roaster B. Both advertise that their morning pour-over takes an average of 4 minutes.

At Roaster A, every cup is served between 3 minutes and 50 seconds and 4 minutes and 10 seconds. At Roaster B, your coffee might arrive in 1 minute, or you might wait 7 minutes. Even though both shops boast an identical average wait time of 4 minutes, Roaster A delivers predictability, while Roaster B gives you high variability and inconsistency.

Modern Practical Explanation

While the median and IQR focus on rank and order, the **Standard Deviation (σ or s)** measures the average distance that values scatter around the arithmetic mean.



Empirical Rule of standard deviation. Source: Iamnee / Getty Images

Variance (s^2): The average squared difference from the mean:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

Standard Deviation (s): The square root of the variance, returning the dispersion back into the original units of measurement:

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

For bell-shaped, symmetrical data, the **Empirical Rule** provides clear baselines:

- ≈ 68% of values fall within $\bar{x} \pm 1s$
- ≈ 95% of values fall within $\bar{x} \pm 2s$
- ≈ 99.7% of values fall within $\bar{x} \pm 3s$

Unlike the IQR, the standard deviation takes every single data point into account, which makes it sensitive to heavy outliers and skewness.

Business Case Study: E-Commerce Delivery Performance

Scenario: *LogiExpress* guarantees same-day and next-day deliveries across two major fulfillment hubs: Hub Alpha and Hub Beta. The VP of Operations wants to evaluate both facilities based on their delivery times (in hours) over a sample of 11 typical orders.

Step-by-Step Calculations & Reasoning

Ordered sample delivery times (in hours):

Hub Alpha: 14, 16, 18, 19, 20, 20, 21, 22, 23, 25, 39

Hub Beta: 18, 19, 19, 20, 20, 20, 21, 21, 22, 22, 23

1. Hub Alpha Metrics

Five-Number Summary:

Sample size: $n = 11$

Min: 14

Q_1 : Lower half is {14, 16, 18, 19, 20} → $Q_1 = 18$

Median (Q_2): 6th value → $Q_2 = 20$

Q_3 : Upper half is {21, 22, 23, 25, 39} → $Q_3 = 23$

IQR: $Q_3 - Q_1 = 23 - 18 = 5$

Outlier Thresholds:

Upper Fence = $Q_3 + 1.5 \times \text{IQR} = 23 + 1.5(5) = 30.5$

Since $39 > 30.5$, **39 is an outlier**. The whisker reaches the maximum non-outlier, which is 25.

Summary: [14, 18, 20, 23, 25] with outlier at 39.

Mean and Standard Deviation:

Sum: $14 + 16 + 18 + 19 + 20 + 20 + 21 + 22 + 23 + 25 + 39 = 237$

Mean: $\bar{x} = \frac{237}{11} \approx 21.55$ hours

Squared deviations:

$(14 - 21.55)^2 + (16 - 21.55)^2 + \dots + (39 - 21.55)^2 \approx 424.73$

Sample Standard Deviation:

$s = \sqrt{\frac{424.73}{10}} \approx 6.52$ hours

2. Hub Beta Metrics

Five-Number Summary:

Min: 18

Q_1 : 19

Median (Q_2): 20

Q_3 : 22

Max: 23

IQR: $22 - 19 = 3$

Upper Fence: $22 + 1.5(3) = 26.5$ (No outliers exist).

Summary: [18, 19, 20, 22, 23]

Mean and Standard Deviation:

Sum: 225

Mean: $\bar{x} = \frac{225}{11} \approx 20.45$ hours

Sample Standard Deviation:

$s \approx 1.51$ hours

Business Interpretation

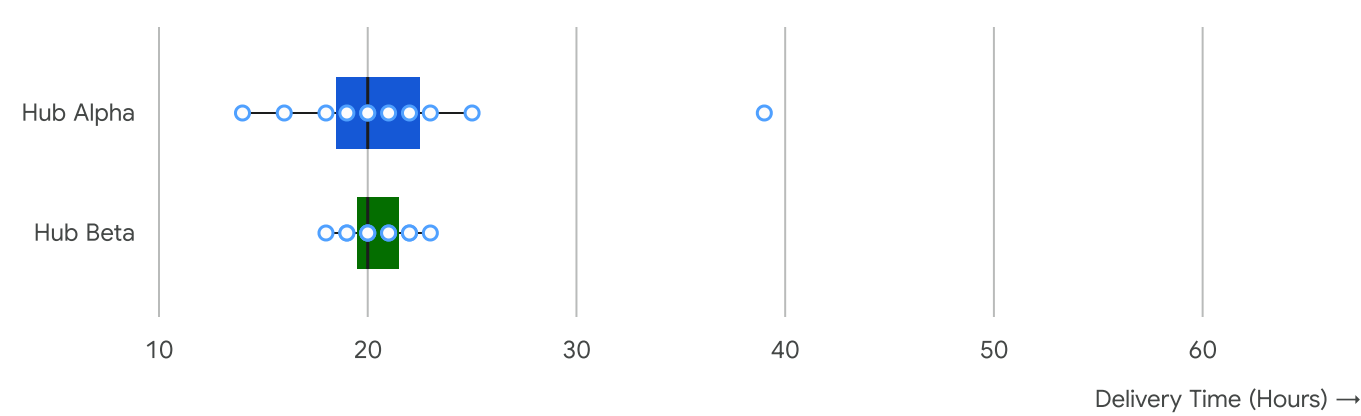
Both hubs show a median delivery time of **20 hours**. However, relying only on central tendency masks key operational risks:

Hub Alpha has a standard deviation of **6.52 hours** and a severe delivery delay outlier at 39 hours (possibly due to driver shortages or vehicle breakdowns), posing a risk to customer satisfaction SLAs.

Hub Beta is consistent and predictable, showing an IQR of only **3 hours** and a standard deviation of **1.51 hours**.

Interactive Comparison Visualizer

Use this interactive tool to test different delivery values, adjust outliers, and inspect how the box plot, quartiles, and standard deviation respond in real time.



Five-Number Summary (Hub Alpha)	
Min (x_0)	14.0 h
Q1 (25%)	18.5 h
Median (Q2)	20.0 h
Q3 (75%)	22.5 h
Max (x_n)	39.0 h

Hub Comparison Gauges

Hub Alpha Std Dev: 6.56

Mean: 21.5 h

Hub Beta Std Dev: 1.51

Mean: 20.5 h

MEDIAN	MEAN	STD DEV	IQR
20.0 h	21.5 h	6.56	4.0 h

Select Hub

Hub Alpha

Max Outlier Value

39

Was this visual helpful?

Yes

No